On Theoretical Justification of Empirically Observed Properties of Financial Markets

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Abstract

We suggest an unconventional physical approach to describe the market pricing process. Using it, we have developed a model which allows not only estimating the volatility of future market prices, but also permits forecasting in what direction, up or down, the price will go. Besides, the model's solution possesses properties consistent with those of real markets: it is fat-tailed, evinces scaling and allows detecting nonlinear dependence ("market memory") in successive price changes. Furthermore, as our model contains no parameters to be tuned on historical market quotes, the consistency takes place under no model calibration, and hence the developed model may not solely produce a yet another confirmation of the known empirical facts about markets, but provide a theoretical justification thereto. Tests of the model's prototype on real financial instruments prove the competence of our approach.

Keyword: fat-tailed probability distributions, scaling of price changes, nonlinear market memory, open systems with non-conserved energy

1 Introduction

By now many important facts about behaviours of financial markets have been established [3, 4, 5]. Among them are the fat-tailed character of distributions of price changes, their scaling, (nearly) zero correlation of price changes, as well as the power character of the correlation of squares thereof. However, so far it is still unclear *why* price changes possess these properties.

Currently one may distinguish two main ways in econophysical studies. The first implies engagement of a predetermined equation from some or other area of physics and subsequent tuning of the equation's parameters on historical market quotes [4, 5]. The second way is agent-based simulation [5]. Although many of the corresponding studies have made essential contributions to the discovery of emprical market laws, we think those models do not provide a theoretical *justification* to the established facts. Besides, in our view, both approaches are vulnerable in the sense that even a tiny debatable supposition underlying a specific model may eventually lead to a result inadequate to market reality. Therefore, efforts should be focused on the development of such an analytical market dynamics model which would reflect the reality without involving arguable suppositions.

2 Underlying Idea

We consider the entire set of market agents as an open mechanical system of two players, the "bull" and the "bear", each of whom is "tugging" the price to his side. New information thrown into the market is likened to the energy introduced into the system. Although the exact relevant "bullish" and "bearish" factors affecting the price are unknown, all we need is to find out in how many times the probability of a price change is greater or less than the probability of the other, independently of the specific market factors resulting in such a change. This is done via the estimation of the amount of the incoming energy: the more energy to income, the less likely the corresponding price change to happen. As the system is treated open, the energy can have any value, i.e. it is not conserved over time [2]. Besides, we regard the price as being formed simultaneously at diverse time scales (e.g., minutes, days, etc.). Enumerating over possible values of the future price change, we determine the least "energy-expensive" and therefore the most probable ones. Such an approach allows developing a model which takes into consideration all incoming information, but makes no debatable a priori assumptions about markets nor requires a posteriori calibration carrying no parameters to be tuned.

3 Results

Mathematically the above-stated idea results in a formalisation of the form

$$Pr(\Delta x) = Pr({}^{j}\Delta x_{next} \mid {}^{j}\Delta x_{prev}). \qquad (1)$$

Here ${}^{j}\Delta x_{prev}$ and ${}^{j}\Delta x_{next}$ are two successive price changes at the scale $j \in \mathbb{N}$, so for each known state

$^{j}\Delta x_{prev}$ the model provides the probability of transition to the state $^{j}\Delta x_{next}$. The model's solution is thus a family of probability distributions.

The developed model possesses the following properties:

1. The probability distributions are asymmetric fat-tailed (Fig. 1) with the degree of "fat-tailedness" $\alpha \approx 0.6 - 0.9$.

2. The distributions' asymptotic ("tailed") symmetry results in zero correlation between price changes. Besides, the correlation of squares of price changes is of a power law [1].

3. Despite the zero correlation, the correlation field between successive price changes is non-random (Fig. 2), which implies nonlinear dependence ("market memory").

4. The distributions' structure is kept unchanged when passing from one scale to another, so the property of scaling holds [1].

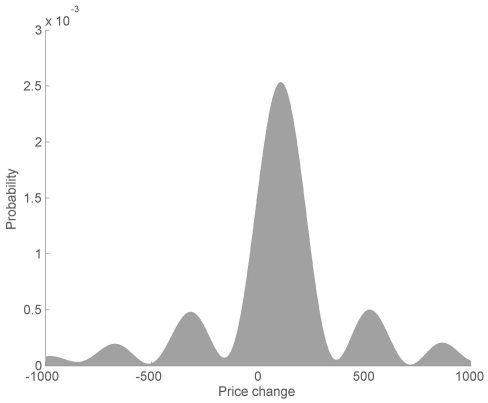


Figure1: An example of the model's solution (shown approximated)

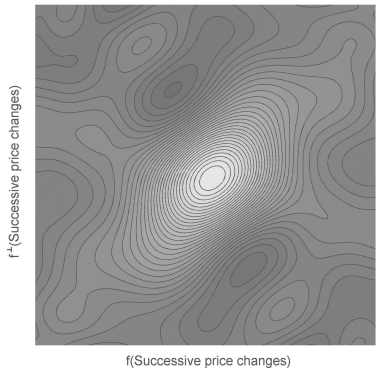


Figure2: Correlation field simulated with the model (shown smoothed and affinely transformed)

Preliminary numerical tests for the currency pair euro/dollar, the oil, the gold, the wheat and the stocks of Google and Apple on diverse time scales demonstrate statistically significant (at the average above 1%) skewnesses of the relative frequency of success when forecasting the direction of future price changes (Fig. 3).

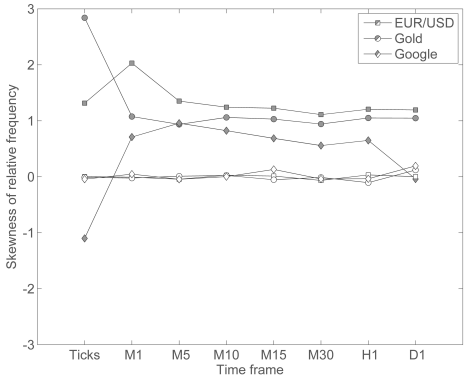


Figure3: Graphs of the relative frequency's skewness ($Fr - 50\%$) to make the right forecast of the direction of future price change for the original (filled markers) and randomly permuted (hollow markers) price changes

4 Conclusion

Alongside with the model's consistency with the empirical observations, this is exactly its analyticity that makes it different from other econophysical models. In other words, instead of an imitation of the real pricing process our model provides a theoretical justification to the properties of real markets. This suggests one may expect a complete pricing model based on the developed prototype to possess a practically apt ability to predict future price changes.

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